



ROBINHOOD CHAIN MAINNET

Protocol White Paper

Forge positions, recurring ETH and cbBTC rewards, and user-triggered buy-and-burn pools.

VERSION 1.1 - 6 OCTOBER 2026

This edition describes the live Robinhood Chain deployment and ownership restrictions confirmed on 6 October 2026. Position and Foundry ownership is renounced; each burner is owned by the restricted daily-rate controller. Only the 1%-10% daily drip remains adjustable.

thefuelforge.com

01 Executive summary

Fuel Forge is a set of smart contracts on Robinhood Chain that uses FUEL in two different ways. A Forge position locks FUEL as principal and optionally burns additional FUEL to increase its power. A separate Foundry NFT permanently burns 100 million FUEL and participates in a recurring cbBTC reward pool. ETH-denominated Forge mint fees fund three ETH reward pools, the Foundry cbBTC pool, three token buy-and-burn pools, and a development wallet.

The protocol does not promise a fixed yield. Rewards depend on actual fee inflows, the number and power of eligible positions or Foundry NFTs, successful market swaps, and someone submitting the necessary settlement transactions. A position's NFT carries its remaining principal and unclaimed reward rights when transferred. The protocol has not received an independent security audit.

Component	Current mainnet setting	Who acts
Forge position	8-1,000 days; up to 5x initial power	Holder creates; holder withdraws after maturity
ETH reward pools	8-, 28-, and 88-day cycles	Anyone can settle; rewards are claimed
Foundry NFT	100 million FUEL burned per NFT; 288-day cbBTC cycles	Holder mints; anyone can settle
Buy-and-burn	FUEL, MORE, PAMP; 10-minute intervals	Someone presses Execute burn

TWO DISTINCT ECONOMIC POSITIONS

Forge principal is locked and may be returned after maturity. Foundry's 100 million FUEL per NFT is permanently burned at mint. An optional Forge power burn is also permanent.

02 Release scope and architecture

This paper covers the Robinhood Chain deployment with chain ID 4663 and the contract addresses in Section 13. Its launch anchor is 26 September 2026 at 17:00 UTC (1:00 PM Eastern Daylight Time). Deadlines remain fixed in UTC, so their displayed local time may shift when daylight saving time changes. The website at thefuelforge.com is an interface to these contracts; the blockchain records the transactions and controls the outcomes. Any future deployment on another chain would have separate contracts, assets, positions, and cycles.

The principal components are: the ForgePosition ERC-721 contract; three PositionRewardVault contracts for ETH; the ForkFoundry ERC-721 contract and its cbBTC conversion adapter; a one-time fee forwarder and immutable FeeRouter; three ForkBurnEngine contracts with swap adapters; and an optional SettlementBatcher. FUEL, MORE, PAMP, WETH, and cbBTC are existing external tokens, not newly issued Fuel Forge tokens.

The mint and burn transactions require the connected wallet to be on Robinhood Chain. Users pay network gas for their own transactions. Burn execution, pool settlement, and expired-position finalization are intentionally user-triggered; there is no required automated keeper. Delayed execution delays the resulting purchase or claim availability, but does not move a pool's scheduled deadline.

03 Forge positions and power

A Forge position is minted by locking a positive amount of FUEL for a whole-number term from 8 through 1,000 days. The creator may also send up to three times the principal to the burn address. The principal and optional burn are transferred in the same position-creation transaction. The burn cannot be recovered. The initial power is computed once at mint:

INITIAL POWER

Power = principal + floor(principal x (termDays - 8) / 992) + burned FUEL. Because the duration bonus is at most the principal and the burn is at most three times the principal, initial power is at most five times principal.

At the shortest term, the duration component is zero. At 1,000 days, it equals the principal. Burned FUEL adds power one-for-one within the 3x cap. The contract checks the term, positive stake, and burn cap. Power is measured in FUEL-sized units for allocation; it is not a separate token or a balance that can be sold.

The owner may transfer a live Forge NFT. A transfer changes who controls the principal and who may receive unclaimed ETH rewards; it does not reset creation time, maturity, burn, or power. There is no early principal withdrawal. A position may be minted before the first cycle begins, and then becomes eligible for a later scheduled close if it has positive power at that close.

04 Maturity, grace, and decay

Maturity is the mint block timestamp plus the selected number of calendar days. Principal and power remain at their full then-current amounts through a seven-day grace period after maturity. Over the following seven days, both decline linearly to zero. The contract rounds remaining principal and power down separately. After fourteen days past maturity, no principal or power remains.

When the owner acts	Principal outcome	Power at future snapshots
Before maturity	Withdrawal unavailable	Initial power
Maturity through day 7 late	100% of remaining principal returned	Initial power until withdrawn
Day 7 through day 14 late	Linearly decreasing amount returned; rest burned	Linearly decreasing until withdrawn
Day 14 late or later	0 returned; all principal burned	0

Withdrawal closes and burns the position NFT. The contract records the final NFT owner so reward claims already earned before closure still have a recipient. At full expiry, anyone can call `finalizeExpired`, which burns the remaining principal and closes the NFT. Neither maturity nor full expiry automatically sends a transaction; a person must act and pay gas. Previously allocated rewards are not reduced by later principal decay or finalization.

05 ETH reward cycles

Three global ETH pools close every 8, 28, and 88 days from the launch anchor. They are funded with fixed shares of each Forge position fee. A position in any allowed term can qualify for any of these pools if it was created strictly before that pool's scheduled deadline, has positive power at that deadline, and was not closed at or before that deadline. Eligibility is based on the scheduled deadline even when settlement occurs later.

Pool	Share of Forge fee	Share of 41% reward bucket	Cycle
8-day ETH	13.12%	32%	8 days
28-day ETH	11.07%	27%	28 days
88-day ETH	8.61%	21%	88 days
Foundry cbBTC	8.20%	20%	288 days

After an ETH pool closes, anyone may process its historical position snapshot in bounded batches. The pool reserves 0.25% of its gross ETH for the callers who finish those batches. The remaining 99.75% is the participant pool. For an eligible token, the claim is $\text{floor}(\text{participantPool} \times \text{tokenPowerAtDeadline} / \text{totalEligiblePower})$. If total eligible power is zero, the participant amount rolls into the then-current cycle instead of being allocated to nonexistent positions.

Claims are pull-based and do not expire in the current contracts. Anyone may trigger a claim, but ETH is sent to the position's current NFT owner, or its recorded final owner after closure. Thus transferring an NFT transfers its unclaimed reward rights. After all eligible claims, indivisible wei dust rolls into the then-current cycle.

Before settlement, the website's pool balances, power shares, and payouts are estimates. A displayed USD reference is informational and is not an on-chain redemption value. The scheduled snapshot and final on-chain state govern the claim.

06 The 288-day Foundry

Foundry is separate from a Forge stake. Each NFT mints by permanently transferring 100 million FUEL to the burn address and paying an ETH mint fee. There is no per-wallet NFT limit in the contract. Foundry NFTs minted into the same 288-day cycle each receive one equal share of that cycle's cbBTC participant pool after settlement. They do not participate in later Foundry cycles unless newly minted for those cycles.

The Foundry mint fee is fixed at 100% of the live quoted ETH value of 100 million FUEL, with no minimum or maximum fee clamp. Ownership is renounced. Of the fee, 1% goes to development and 99% buys cbBTC. The Foundry also receives 8.20% of Forge position fees through the router. Gas is additional.

At the scheduled cycle close, anyone may settle the Foundry pool. The settler receives 0.25% of gross cbBTC; the remainder is divided equally among eligible NFTs. Claims remain available after settlement. A Foundry NFT can be transferred, and the current owner receives its unclaimed reward. If no NFTs are eligible, or if division leaves smallest-unit dust after all claims, the remaining amount rolls into the then-current Foundry cycle.

CONVERSION CAN FAIL

The Foundry conversion cap is fixed at 2 ETH per swap after ownership renunciation. If a swap fails, the contract can retry smaller protected swaps inside the same wallet transaction, subject to 32 completed swaps and 64 attempts. Quote failures, insufficient output, or exhausted limits revert the entire mint or fee conversion; no FUEL is burned on a reverted mint.

07 Fee policy and routing

The Forge entry fee is fixed at 100% of the quoted ETH value of principal FUEL only. Optional burn and term do not increase it. Forge and Foundry fee bounds are disabled and both owners are the zero address, so their fee policies cannot be changed through the former owner setters.

Destination	Share of each Forge fee	Role
8-day ETH pool	13.12%	Position rewards
28-day ETH pool	11.07%	Position rewards
88-day ETH pool	8.61%	Position rewards
Foundry	8.20%	cbBTC purchases and rewards
FUEL burner	33.00%	Buy FUEL and burn
MORE burner	20.00%	Buy MORE and burn
PAMP burner	5.00%	Buy PAMP and burn
Development	1.00% nominal	Development wallet; also receives indivisible wei dust

The existing FeeRouter's shares and recipients are immutable. The Forge contract sends fees through a one-time configured forwarder to that router. Routing happens atomically: if a required receiver transfer fails, the mint reverts. The Foundry mint-fee path is separate: its 1% development share and 99% cbBTC-conversion share are not passed through this router.

FIXED MINT FEES

Forge and Foundry ownership is renounced. Their current 100% fee percentages and disabled bounds cannot be changed through an owner call. Values still stored in the minimum/maximum getters are inactive because feeBoundsEnabled is false.

08 Buy-and-burn execution

The FUEL, MORE, and PAMP burn engines receive ETH from Forge fees. Each currently makes 1% of its pool balance available per day through 144 scheduled 10-minute intervals. The owner may set each engine's daily rate between 1% and 10%. Unexecuted intervals accumulate rather than vanishing. A later execution may cover multiple elapsed intervals, but each transaction is limited by that engine's maximum ETH per execution.

All three burn engines have a fixed 1 ETH gross execution cap and 10% slippage limit under the restricted-controller ownership arrangement. The controller exposes only daily-rate changes, so its manager cannot call the underlying cap or slippage setters. A successful execution pays 1.5% of gross ETH to the caller and swaps the other 98.5% for tokens sent to the dead address. Gas may exceed the caller reward.

The displayed 'next burn' is an estimate based on current balance, elapsed intervals, and the cap. It is not a promise that every press succeeds. A stale or invalid reference quote, a failed router swap, a price move outside the applicable guard, or insufficient output causes the transaction to revert. The 10% slippage tolerance is a minimum-output check and does not override an earlier quote-integrity check. The burn then waits for a later successful user-triggered execution; there is no automatic keeper.

In simplified terms, newly executable ETH is the current pool balance multiplied by accumulated interval weight divided by $144 \times 10,000$, plus any previously executable amount left by an earlier cap. The result cannot exceed the contract balance. An owner rate change checkpoints the weight already earned at the prior rate; a later execution uses the combined amount and is still capped per transaction.

09 Price references and market execution

Fees and swaps depend on external liquidity pools. Forge and Foundry calculate ETH fees from a live FUEL price oracle. The mint oracle first attempts a checked FUEL quote; on certain volatility or cross-pool disagreement errors it falls back to a live spot quote from the FUEL pool. That fallback helps minting remain available but is less resistant to a temporarily manipulated thin pool. Other quote failures can still prevent a mint.

The FUEL and MORE burn paths compare observations from more than one pool. PAMP does not have a comparable independent reference in the deployed configuration; it relies on its available pool's guarded quote. cbBTC conversion relies on its WETH/cbBTC pool quote and minimum-output protection. These are pool-based safeguards, not a guarantee of an independent fair market price. Small pools can be manipulated, become stale, or lack enough depth at the moment of execution.

For burn and Foundry conversions, the reference quote must be valid and no more than 15 minutes old. The fixed slippage parameter cannot exceed 10%. A failed swap reverts atomically instead of silently accepting fewer tokens. The protocol cannot promise that every burn, mint, or conversion will complete during adverse market conditions; making that promise would require weakening current protections or changing deployment architecture.

PRICE SOURCE TRADEOFF

The owner accepted thin-pool price references for launch. A cross-pool comparison is useful but does not make two correlated or shallow pools fully independent. FUEL fee spot fallback and PAMP's single-pool reference expose distinct manipulation and availability risks.

10 Governance, ownership, and operations

The daily-rate manager and development recipient is `0x02A0d741FBaebC03A8f0d1A85670bf1CA8C15fA9`. Burn engines are permanently owned by `DailyBurnRateController` at `0xCae88Ca5A86F3eCB94673229C9EDd19e31274fb3`. It exposes only each registered engine's 1%-10% daily drip setter and has no ownership-transfer, generic-call, upgrade or manager-replacement function. Forge and Foundry ownership is renounced. Routes, allocations, fees, swap caps and slippage cannot be changed by this manager.

The inspected contracts do not expose an owner function to withdraw locked Forge principal or allocated participant rewards. This is a code-level observation, not a security guarantee. The current system is unaudited. Users should inspect the addresses and current parameters on-chain because the daily drip can change without this PDF changing.

Anyone can submit an eligible burn, settle a closed reward cycle, claim a reward, or finalize a fully expired Forge position. Public submission creates no automated schedule: if no one sends a transaction, that action waits. The optional settlement helper can group up to 20 items and at most 60 Forge-position visits in one transaction; larger snapshots use the per-pool batch function. Every submitted transaction requires gas and may revert.

11 Worked examples

Forge power

Suppose a user locks 100 million FUEL for 1,000 days and permanently burns another 300 million FUEL. Base power is 100 million; the duration bonus is 100 million; the burn bonus is 300 million. Initial power is 500 million, or 5x the principal. Only the 100 million locked principal is potentially returnable after maturity.

ETH payout

Suppose one ETH pool closes with 10 ETH gross and all eligible positions total 10 billion power. Settlement reserves 0.025 ETH (0.25%) for callers, leaving 9.975 ETH. A 500 million-power position has a 5% snapshot share and can claim 0.49875 ETH before smallest-unit rounding. This is hypothetical; actual pool funding and total power are unknown in advance.

Foundry mint and reward

If 100 million FUEL is quoted at 1 ETH, the fixed 100% rule charges 1 ETH for a Foundry NFT: 0.01 ETH goes to development and 0.99 ETH purchases cbBTC, subject to execution. If its mint cycle later has 0.1 cbBTC gross and four eligible NFTs, the 0.25% settlement incentive leaves 0.0249375 cbBTC per NFT before rounding. These are hypothetical values, not forecasts.

Burn execution

If an engine has 0.001 ETH executable below its fixed 1 ETH cap, a successful execution pays 0.000015 ETH to the caller and spends 0.000985 ETH on the token purchase. The number of tokens actually burned depends on the realized swap price; an unsuccessful swap pays no caller reward and makes no burn.

12 Material risks and limitations

Risk	Practical effect
Smart-contract and integration errors	Loss, trapped assets, wrong accounting, or failed execution. No independent audit has been performed.
Thin liquidity and price manipulation	Mint fees can reflect a distorted FUEL quote; burns or cbBTC conversion may revert or execute at an unfavorable permitted price.
Asset and counterparty risk	FUEL, MORE, PAMP, cbBTC, ETH, and NFTs may lose value. cbBTC also carries token issuer/custody and bridge or chain risks.
Irreversible burns and delayed withdrawal	Optional Forge burns and Foundry burns cannot be recovered. Forge principal cannot exit early and reaches zero after the late-decay period.
Owner powers	Only the daily burn drip can change, within 1%-10%. Renounced ownership and restricted controllers remove the other former owner settings. Contract and market risks remain.
Execution dependence	Rewards must be settled and claimed; burns must be triggered. Gas costs, RPC problems, and failed quotes can delay actions.
No fixed yield	Fees and rewards depend on real usage and competing eligible positions. Buy-and-burn does not guarantee token price support.

This document describes mechanisms and risks; it is not investment advice. No APY, token appreciation, or principal recovery after the defined decay period is promised. A wallet confirmation should be reviewed carefully before signing.

How these risks meet in practice

A burn interval can be ready while its swap fails a quote guard. Funds remain for a later execution; a reverted transaction still costs gas. Entry spot fallback can keep minting available during selected price disagreements, but may value FUEL at a briefly distorted pool price. The 100% fee has no active floor or cap and cannot be changed by an owner.

Settlement callers should compare likely caller compensation with their gas cost. NFT holders should track their own maturity and seven-day grace period; no automation withdraws principal for them. The live app is helpful for monitoring, but the contract timestamps and transaction receipts are decisive.

13 Mainnet contract registry

The addresses below come from the live site's deployment manifest and were checked against public contract getter readings on 6 October 2026. Check the block explorer for current code and state before using an address. Earlier testnet, fork, and superseded mainnet contracts are outside this registry.

Contract	Robinhood Chain address
Forge Position NFT	0x316a858567426AcD376E242b90CeD3742101A58b
Foundry NFT	0x6A6657010862934AC701eDf1E1f246018250953e
8-day ETH vault	0x9F45B37540bCc401Ff5C0d0Da02d21E87bE6fAD7
28-day ETH vault	0xee0fcF6C7a7c50B281eF69A304ccF00903483a6B
88-day ETH vault	0x7ABFD5402fC7A0ec67a750990afe0Efb921FAA40
FUEL burner	0x20c6D511Cb4862e30B2d28060DCAEfc86f2DC876
MORE burner	0x5Df3D7abA3dA3F6a2d89a7bA689294F6312A85DF
PAMP burner	0x70FfdAf9277796711de29A5A4c82dDd5EACaB3d7
Fee router	0x60be7C8F85f3A67cF6321444CB960F6605EF9F62
Fee forwarder	0xd2A61229113Ec1e629d25b1EB6F3ec5708e12576
Settlement batcher	0x5b11A4368C6ed843F0625284184286D0dd34F3C6
Daily-drip controller	0xCae88Ca5A86F3eCB94673229C9EDd19e31274fb3

External token addresses used by this deployment

Token	Address
FUEL	0xe60C1F5d9bA7f62a392a78472a3Ab83DD62467A3
MORE	0xc0F1A40512114b25cc1F30b5DF0bb48691405555
PAMP	0x64D1472d061a6B4a0ebE4B31Ad30f3774df219a6
cbBTC	0xCEC185eB182c47d1bA1EFc84e6959e18cd620Be4

14 Verification and versioning

Version 1.1 was prepared from the deployed manifest, the Solidity implementation used for the mainnet candidate and mint-availability replacement, and read-only public-RPC calls to the deployed Forge, Foundry, fee router, reward vaults, and burn engines on 6 October 2026. The public readings confirmed the day and cycle durations, launch timestamp, current fee settings, burn intervals and caps, owner, token references, and router destinations. The figures in Sections 3-11 were cross-checked against the named contract functions.

This revision records the final 100% fees without clamps, renounced position/Foundry ownership, restricted daily-drip controller, 1 ETH burn caps and 2 ETH Foundry conversion cap. The daily drip and market conditions can change; the other former owner controls are inaccessible.

Primary references: the deployed contracts at <https://robinhoodchain.blockscout.com/> ; the live interface and manifest at <https://thefuelforge.com/> and <https://thefuelforge.com/mainnet-deployment.json> ; and the Solidity source files ForgePosition.sol, ForgeMath.sol, FeePolicy.sol, PositionRewardVault.sol, ForkFoundry.sol, FeeRouter.sol, ForkBurnEngine.sol, ForkSwapAdapter.sol, and SettlementBatcher.sol in the project source used for this deployment.

Glossary

Term	Meaning in this paper
Principal	FUEL locked in a Forge position and subject to maturity and late decay.
Power	A non-transferable accounting weight attached to the Forge NFT; not a separate token.
Cycle	A globally scheduled period for collecting and allocating one reward pool.
Snapshot	Position power at the scheduled deadline, even if settlement is later.
Settlement	A submitted transaction that closes and allocates a completed cycle.
Gross / participant pool	Gross is the full cycle amount; the participant pool is after the 0.25% caller allocation.
Slippage	Permitted shortfall from a reference output quote; it does not bypass other quote checks.
Burn cap	The fixed maximum gross ETH spent by one burn execution.

PUBLICATION STATUS

Published protocol documentation dated 6 October 2026. It is not an independent security audit or legal review. Check live on-chain parameters before taking action.